

Technical proof summary: why IEEE-14 does not pin all four GFM after the SG trip

Evidence from report_ieee14_switching_decision_basis_en — every number read from a named artefact

When the SG at bus 1 trips, four dual-mode GFL/GFM converters remain at buses 2, 3, 6 and 8. Why is the forming set not pinned to all four from start to finish, but chosen adaptively at one or two? There are two measured proofs.

1 Proof 1: SCR does not respond to the forming count

The project defines (+stability/ibr_scr_metrics.m)

$$\text{SCR}_i = \frac{|V_i|^2}{|Z_{th,ii}| S_{\text{rated},i}}, \quad Z_{th,ii} = [\mathbf{Y}^{-1} \mathbf{e}_i]_i, \quad (1)$$

where $\mathbf{Y}$ is built from branch data (including tap and phase shift) plus bus shunts, *excluding* load admittance, with reference buses grounded. No term in $\mathbf{Y}$ depends on how many converters are forming. [measured]

Window	n	min	mean	max	< 3
pre-trip (SG online)	168	2.0892	2.3439	2.4803	100.00%
islanded (SG offline)	2760	0.6501	2.3015	2.8147	100.00%
whole run, 4 converters	14716	0.2944	2.5271	6.6726	98.61%

Table 1: Per-sample SCR at the converter buses. The islanded window averages 2.3015 and sits below the WECC strong-grid floor of 3 for 100.00% of its samples (98.61% over the whole run). The minimum 0.2944 occurs at the bus-9 fault, not at the machine trip. [measured]

The first two rows carry the point: the window mean moves from 2.3439 to 2.3015. The machine trip barely moves SCR, because Z_{th} does not know the machine exists. What changes is *which sources sit behind the bus*, not the impedance.

- Promoting a converter cannot stiffen the network.** The impedance term of (1) is built from branch data, bus shunts and reference-bus grounding only, so adding a former cannot change Z_{th} at all. SCR can move only through the measured numerator $|V_i|^2$, never through the count. (This does not claim SCR never moves.)
- The problem changes kind, not degree.** REGC_A screens a current-source plant because a PLL needs a stiff bus to lock to. With all four forming there is no PLL left to protect, and the question becomes whether four voltage sources can share current on a high-impedance network — a question this network answers badly (Proof 2).
- Scope that must travel with the number.** In ibr_scr_metrics.m:266--276 the gate is forced to profile not_applicable_full_state_source_model for eecon49_dual, giving eligible_for_scr = false and pass = true. The figures above are therefore a per-sample published diagnostic, not an active gate. [measured] The network-invariance conclusion does not depend on this: it is a property of the equation, not of the gate's state.

2 Proof 2: damping margin and dynamic failure

The project enumerates all $2^4 - 1 = 15$ forming sets, solves each equilibrium and assembles its full-KCL spectrum. **Only 7 of the 15 are admissible**, and **all four three-unit sets have no equilibrium at all** (the coupled Newton does not converge; residuals 3.279, 3.605, 3.561, 3.346×10^2). What actually gates is equilibrium existence + device limits + full-KCL SSSA with a damping-ratio floor $\zeta_{\min} = 0.05$ on every mode of the decision spectrum, evaluated at three frozen finite-difference steps. $\gamma_{\text{req}} = 0.1$ rad/s is the *ordering key* among already-admissible candidates only, not a gate — the margin may be negative while a set is admissible.

Forming set	ω (rad/s)	ζ_{worst}	Outcome
{5}, {4}, {3}, {2} (one)	−0.1554 to −0.3280	0.2569–0.2676	admissible
{2, 4}, {3, 4} (two)	−0.5617, −0.5681	0.2549, 0.2560	admissible (most damped)
{2, 3, 4}, {2, 3, 5}, {2, 4, 5}, {3, 4, 5}	—	—	no equilibrium
{2, 3, 4, 5} (four)	−0.4829	0.1223	admissible, least damped

Table 2: ζ_{worst} for the other six admissible sets (all four singletons, plus {2, 4} and {3, 4}) spans 0.2549–0.2676; the seventh, the four-unit set, sits at **0.1223** — **less than half** — even though it still clears the 0.05 floor. [measured]

The 0.05 floor is a minimum, not a target. The four-unit set *passes* the gate yet is the worst of the six survivors, and going from two units to three is not a compromise — it lands where the equations have no solution at all.

Arm	horizon	samples	reclose	termination
locked_gf1	20.0000 s	43	—	noVoltageFormingSource
pinned_gfm4	25.4852 s	837	never requested	adaptiveDtMin
pinned_gfm2	250.0000 s	1761	154.0244	SUCCESS
adaptive	250.0000 s	3679	159.2519	SUCCESS

Table 3: Six arms from one shared option set. `pinned_gfm4` dies at 25.4852 s — **it survives 5.4852 s after the SG trip**, before the load step and before any fault. **[measured]** The table also shows that *pinning* is not the defect: `pinned_gfm2` pins two and reaches 250 s. The evidence condemns the count of four, not the pinned mode.

The terminal state of `pinned_gfm4` is not a gradual loss of damping but **frequency splitting**: in its last accepted sample IBR2 runs at 64.9947 Hz while IBR3, IBR6 and IBR8 run at 59.1383, 59.1809 and 59.1946 Hz — **a 5.86 Hz split**, one converter having run away from the other three. The step controller then reported **398 successive Newton non-convergences** and terminated at $\Delta t_{\min} = 6.104 \times 10^{-6}$ s. **[measured]**

Why an admissible set can still die: the acceptance gate is a statement about an *equilibrium*. It is proved by linearising about the solved operating point and says nothing about the trajectory that arrives there. The defect record measured that gap by re-running the same inputs from two different initial device states.

Arm	initial device state	outcome	excursion
ARM A	the state the production run actually arrives with	SLIPS	4358°
ARM B	the certified equilibrium state of the same configuration	SYNC	6.6°

Table 4: The same configuration, the same inputs, two initial states. The carrying coordinate is the outer voltage-loop integrator gfm_xi_Vd/xi_Vq of IBR2: 0.144253 on arrival against 0.199658 at the certified equilibrium — a **28%** displacement in a coordinate no bus quantity reflects. **[measured]**

Scope of that table. The ARM A/B experiment is measured on a different all-four transition from the one that kills `pinned_gfm4`: it is the load-step commit at $t = 53.4025$ s in the pre-certificate baseline, not the death at 25.4852 s. Both are the same *class* of failure — an admitting equilibrium whose arriving state lies outside its basin of attraction — but they are not the same event, and the record does not root-cause the pinned arm’s death. Structurally: the three-unit sets fail the *weaker* test (no equilibrium exists), while the four-unit set passes the equilibrium test and fails the test of the *trajectory* — a failure no eigenvalue check can see.

3 Executive conclusion

- **“More GFM is NOT safer” — in both directions.** The lower bound is measured and hard: `locked_gf1`, with no former at all, dies at $t = 20.0000$ s, the trip instant itself. The upper bound is measured too, and it is not four: `pinned_gfm4` dies at 25.4852 s.
- **Adding GFM does not stiffen the grid.** The impedance term of the SCR equation is built from branch data, shunts and reference-bus grounding only; the count cannot change it. The islanded window averages 2.3015 and sits below the threshold of 3 for 100.00% of the islanded time (98.61% over the whole run). Pinning all four therefore does not fix the weak grid; it moves the problem to four voltage sources sharing current with each other.
- **The right count is the smallest one still admissible.** Two-unit sets are *better damped* than the four-unit set by nearly a factor of two (ζ_{worst} 0.2549–0.2676 against 0.1223), and every three-unit set has no equilibrium at all. The only exit consistent with the evidence is to start at one and grow to two when it is genuinely required — which is what a *minimum-cardinality admissible set* does.
- **The counter-evidence belongs in the same answer.** The defect record contains a measured case where the four-unit configuration rides a 2.0 pu load deficit that the two-unit configuration cannot (the defect record’s load-severity sweep, at a different load level — a separate measurement from the delivered run and from the frozen base-load table). The defensible conclusion is therefore not “fewer is better” but: **the count is not a free parameter and stability is not monotone in it**. It has to be chosen per configuration, from the certified margin and from the arrival.